

Bounds for $Q_k^{\text{bag},\infty}$ under Common Loading

Summary note

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1 Objective

The goal is to bound the infinite-bagged RSM population risk

$$Q_k^{\text{bag},\infty}$$

using only quantities that are already available in the common-loading framework: the average-subset risk Q_k^{sub} , the equal-weight risk Q^{ave} , and the optimal risk Q^{opt} . The main practical question is whether these bounds can provide a usable functional form for $Q_k^{\text{bag},\infty}$ as a function of the subset size k .

The key conclusion is:

$$Q^{\text{opt}} + \frac{\left[\frac{k}{m}(Q_k^{\text{sub}} - q) - (Q^{\text{opt}} - q)\right]^2}{Q^{\text{ave}} - Q^{\text{opt}}} \leq Q_k^{\text{bag},\infty} \leq \min\{Q_k^{\text{sub}}, Q^{\text{ave}}\}.$$

After plugging the common-loading closed forms, this becomes

$$q + \frac{B}{m} + \frac{[k(Q_k^{\text{sub}} - q) - B]^2}{mB(H-1)} \leq Q_k^{\text{bag},\infty} \leq \min\left\{Q_k^{\text{sub}}, q + \frac{BH}{m}\right\}.$$

The first-order approximation $Q_k^{\text{sub}} \simeq q + B/k$ makes the numerator of the lower bound vanish. Therefore, the first-order Q_k^{sub} approximation gives only the trivial lower bound $Q_k^{\text{bag},\infty} \geq Q^{\text{opt}}$. A nontrivial lower bound appears only after retaining the second-order term in Q_k^{sub} .

2 Setup and known quantities

Assume the common-loading heteroskedastic forecast-error covariance structure

$$\Sigma_e = q\mathbf{t}_m\mathbf{t}_m' + D, \quad D = \text{diag}(\sigma_1^2, \dots, \sigma_m^2), \quad \tau_i = \sigma_i^{-2} > 0.$$

For a subset S of size k , write

$$A_S = \sum_{i \in S} \tau_i, \quad A_m = \sum_{i=1}^m \tau_i = m\bar{\tau}, \quad B = \frac{1}{\bar{\tau}}.$$

Under common loading, the subset-optimal weight embedded into the full m -dimensional vector is

$$v_{S,i} = \mathbf{1}\{i \in S\} \frac{\tau_i}{A_S}.$$

Therefore the infinite-bagged weight is

$$\bar{v}_{i,k} = \mathbb{E}_S[v_{S,i}],$$

and the infinite-bagged population risk is

$$Q_k^{\text{bag},\infty} = q + \sum_{i=1}^m \frac{\bar{v}_{i,k}^2}{\tau_i}.$$

The benchmark risks are

$$Q^{\text{opt}} = q + \frac{1}{A_m} = q + \frac{B}{m},$$

and

$$Q^{\text{ave}} = q + \frac{1}{m^2} \sum_{i=1}^m \tau_i^{-1} = q + \frac{BH}{m},$$

where

$$H = \bar{\sigma}^2 \bar{\tau} = \left(\frac{1}{m} \sum_{i=1}^m \tau_i^{-1} \right) \left(\frac{1}{m} \sum_{i=1}^m \tau_i \right) \geq 1.$$

Finally, the average-subset risk is

$$Q_k^{\text{sub}} = \mathbb{E}_S \left[q + \frac{1}{A_S} \right] = q + \mathbb{E}_S \left[\frac{1}{A_S} \right].$$

3 Upper bounds

There are two useful upper bounds.

3.1 Jensen upper bound

Since $v' \Sigma_e v$ is convex in v ,

$$Q_k^{\text{bag},\infty} = (\mathbb{E}_S[v_S])' \Sigma_e (\mathbb{E}_S[v_S]) \leq \mathbb{E}_S[v_S' \Sigma_e v_S] = Q_k^{\text{sub}}.$$

This gives

$$\boxed{Q_k^{\text{bag},\infty} \leq Q_k^{\text{sub}}.}$$

3.2 Equal-weight upper bound

In the common-loading setup, the population bracketing result gives

$$Q^{\text{opt}} \leq Q_k^{\text{bag},\infty} \leq Q^{\text{ave}}.$$

The endpoint intuition is that $k = 1$ gives equal weighting and $k = m$ gives the full optimal weights:

$$Q_1^{\text{bag},\infty} = Q^{\text{ave}}, \quad Q_m^{\text{bag},\infty} = Q^{\text{opt}}.$$

Combining this with Jensen gives the sharper upper bound

$$\boxed{Q_k^{\text{bag},\infty} \leq \min\{Q_k^{\text{sub}}, Q^{\text{ave}}\}.$$

After plugging $Q^{\text{ave}} = q + BH/m$,

$$\boxed{Q_k^{\text{bag},\infty} \leq \min \left\{ Q_k^{\text{sub}}, q + \frac{BH}{m} \right\}.$$

4 Lower bounds

4.1 Trivial lower bound

Since Q^{opt} is the population minimum over all sum-to-one weights,

$$Q_k^{\text{bag},\infty} \geq Q^{\text{opt}} = q + \frac{B}{m}.$$

Equivalently, define the excess bagging error

$$R_k^{\text{bag}} = Q_k^{\text{bag},\infty} - Q^{\text{opt}}.$$

Then

$$R_k^{\text{bag}} \geq 0.$$

4.2 Projection/Cauchy lower bound

Define the weighted inner product

$$\langle a, b \rangle_\tau = \sum_{i=1}^m \frac{a_i b_i}{\tau_i}, \quad \|a\|_\tau^2 = \langle a, a \rangle_\tau.$$

Let

$$\delta_k = \bar{v}_k - w^{\text{opt}}, \quad \eta = w^{\text{ave}} - w^{\text{opt}}.$$

Then

$$Q_k^{\text{bag},\infty} - Q^{\text{opt}} = \|\delta_k\|_\tau^2, \quad Q^{\text{ave}} - Q^{\text{opt}} = \|\eta\|_\tau^2.$$

By Cauchy-Schwarz,

$$\|\delta_k\|_\tau^2 \geq \frac{\langle \delta_k, \eta \rangle_\tau^2}{\|\eta\|_\tau^2}.$$

It remains to compute the projection term. Since

$$v_{S,i} = \mathbf{1}\{i \in S\} \frac{\tau_i}{A_S},$$

we have

$$\sum_{i=1}^m \frac{v_{S,i}}{\tau_i} = \sum_{i \in S} \frac{1}{A_S} = \frac{k}{A_S}.$$

Taking expectations,

$$\sum_{i=1}^m \frac{\bar{v}_{i,k}}{\tau_i} = k \mathbb{E}_S \left[\frac{1}{A_S} \right] = k(Q_k^{\text{sub}} - q).$$

Therefore,

$$\langle \bar{v}_k, w^{\text{ave}} \rangle_\tau = \frac{1}{m} \sum_{i=1}^m \frac{\bar{v}_{i,k}}{\tau_i} = \frac{k}{m} (Q_k^{\text{sub}} - q).$$

Also,

$$\langle \bar{v}_k, w^{\text{opt}} \rangle_\tau = Q^{\text{opt}} - q,$$

because $w_i^{\text{opt}}/\tau_i = 1/A_m$ and $\sum_i \bar{v}_{i,k} = 1$. Hence

$$\langle \delta_k, \eta \rangle_\tau = \frac{k}{m}(Q_k^{\text{sub}} - q) - (Q^{\text{opt}} - q).$$

Thus

$$Q_k^{\text{bag},\infty} \geq Q^{\text{opt}} + \frac{\left[\frac{k}{m}(Q_k^{\text{sub}} - q) - (Q^{\text{opt}} - q)\right]^2}{Q^{\text{ave}} - Q^{\text{opt}}}.$$

This lower bound is exact as an inequality. It is sharper than Q^{opt} whenever the projection term is nonzero.

5 Plugging the common-loading values

Using

$$Q^{\text{opt}} = q + \frac{B}{m}, \quad Q^{\text{ave}} = q + \frac{BH}{m},$$

we get

$$Q^{\text{ave}} - Q^{\text{opt}} = \frac{B(H-1)}{m}.$$

Also,

$$Q^{\text{opt}} - q = \frac{B}{m}.$$

Therefore the projection lower bound becomes

$$Q_k^{\text{bag},\infty} \geq q + \frac{B}{m} + \frac{\left[k(Q_k^{\text{sub}} - q) - B\right]^2}{mB(H-1)}.$$

Combining lower and upper bounds yields

$$q + \frac{B}{m} + \frac{\left[k(Q_k^{\text{sub}} - q) - B\right]^2}{mB(H-1)} \leq Q_k^{\text{bag},\infty} \leq \min \left\{ Q_k^{\text{sub}}, q + \frac{BH}{m} \right\}.$$

This is the most compact bound using the known population objects.

6 What happens after plugging the first-order Q_k^{sub} approximation?

Let

$$\Delta_k := Q_k^{\text{sub}} - q - \frac{B}{k}.$$

Then

$$k(Q_k^{\text{sub}} - q) - B = k\left(\frac{B}{k} + \Delta_k\right) - B = k\Delta_k.$$

Thus the lower bound can be rewritten as

$$Q_k^{\text{bag},\infty} \geq q + \frac{B}{m} + \frac{k^2 \Delta_k^2}{mB(H-1)}.$$

This expression makes the cancellation transparent.

If one uses only the first-order approximation

$$Q_k^{\text{sub}} \approx q + \frac{B}{k},$$

then

$$\Delta_k \approx 0,$$

so the lower bound collapses to

$$Q_k^{\text{bag},\infty} \gtrsim q + \frac{B}{m} = Q^{\text{opt}}.$$

This is not a contradiction. It means that the first-order term of Q_k^{sub} does not contain enough information to detect how far the bagged weights are from the optimal weights. The first-order term cancels exactly in the projection lower bound.

7 Second-order Q_k^{sub} gives a nontrivial lower bound

The second-order average-subset approximation under common loading is

$$Q_k^{\text{sub}} \approx q + \frac{B}{k} + Bv_\theta \frac{m-k}{k^2(m-1)},$$

where

$$v_\theta = \frac{v_\tau}{\bar{\tau}^2} = \frac{1}{m} \sum_{i=1}^m \left(\frac{\tau_i}{\bar{\tau}} - 1 \right)^2.$$

Therefore

$$\Delta_k \approx Bv_\theta \frac{m-k}{k^2(m-1)}.$$

Plugging this into the lower bound gives

$$\frac{k^2 \Delta_k^2}{mB(H-1)} \approx \frac{B}{m} \frac{v_\theta^2}{H-1} \left(\frac{m-k}{k(m-1)} \right)^2.$$

Hence

$$Q_k^{\text{bag},\infty} \gtrsim q + \frac{B}{m} + \frac{B}{m} \frac{v_\theta^2}{H-1} \left(\frac{m-k}{k(m-1)} \right)^2.$$

The corresponding two-sided approximate bracket is

$$q + \frac{B}{m} + \frac{B}{m} \frac{v_\theta^2}{H-1} \left(\frac{m-k}{k(m-1)} \right)^2 \lesssim Q_k^{\text{bag},\infty} \leq \min \left\{ q + \frac{B}{k} + Bv_\theta \frac{m-k}{k^2(m-1)}, q + \frac{BH}{m} \right\}.$$

This lower bound has the same qualitative $1/k^2$ shape as the candidate form for the bagged approximation error.

8 Interpretation

1. The raw lower bound $Q_k^{\text{bag},\infty} \geq Q^{\text{opt}}$ is always true but too loose.
2. A sharper lower bound can be obtained by projecting the bagged weight error onto the equal-weight-versus-optimal direction.
3. The resulting lower bound depends on the de-commoned quantity $Q_k^{\text{sub}} - q$.
4. Plugging the first-order approximation $Q_k^{\text{sub}} \approx q + B/k$ makes the projection term cancel exactly. Hence first-order Q_k^{sub} gives no nontrivial lower bound beyond Q^{opt} .
5. To obtain a nontrivial lower bound, one must keep the second-order term in Q_k^{sub} .
6. With the second-order term, the lower bound for the excess bagged risk is of order $1/k^2$:

$$Q_k^{\text{bag},\infty} - Q^{\text{opt}} \gtrsim \frac{B}{m} \frac{v_\theta^2}{H-1} \left(\frac{m-k}{k(m-1)} \right)^2.$$

7. The upper bound $Q_k^{\text{bag},\infty} \leq \min\{Q_k^{\text{sub}}, Q^{\text{ave}}\}$ is useful but, by itself, does not identify the exact rate of $Q_k^{\text{bag},\infty} - Q^{\text{opt}}$.

9 Implication for choosing k

The bounds imply that a robust loss bracket for infinite-bagged RSM is

$$(\sigma_\varepsilon^2 + Q_k^{\text{LB}}) I(T, k) \leq L_k^{\text{bag},\infty} \leq (\sigma_\varepsilon^2 + Q_k^{\text{UB}}) I(T, k),$$

where

$$I(T, k) = \frac{T-1}{T-k},$$

$$Q_k^{\text{LB}} = q + \frac{B}{m} + \frac{[k(Q_k^{\text{sub}} - q) - B]^2}{mB(H-1)},$$

and

$$Q_k^{\text{UB}} = \min \left\{ Q_k^{\text{sub}}, q + \frac{BH}{m} \right\}.$$

If one wants a closed-form approximation rather than a bound, additional structure is still needed: finite precision types, concentration of subset precision sums, Gamma precision, lognormal precision, or another distributional restriction. The bounds alone narrow the feasible region for $Q_k^{\text{bag},\infty}$, but they do not fully identify it.

10 Caveat: what if q is not separately known?

The sharper lower bound requires de-commoned risks such as $Q_k^{\text{sub}} - q$ and $Q^{\text{opt}} - q$. If q is not known or separately identified, and only the raw scalars Q_k^{sub} , Q^{ave} , and Q^{opt} are available, then the only universally safe lower bound is

$$Q_k^{\text{bag},\infty} \geq Q^{\text{opt}}.$$

A nontrivial lower bound requires either knowing q , knowing equivalent common-loading primitives such as (B, H, q) , or imposing an additional normalization/distributional assumption that identifies the de-commoned components.